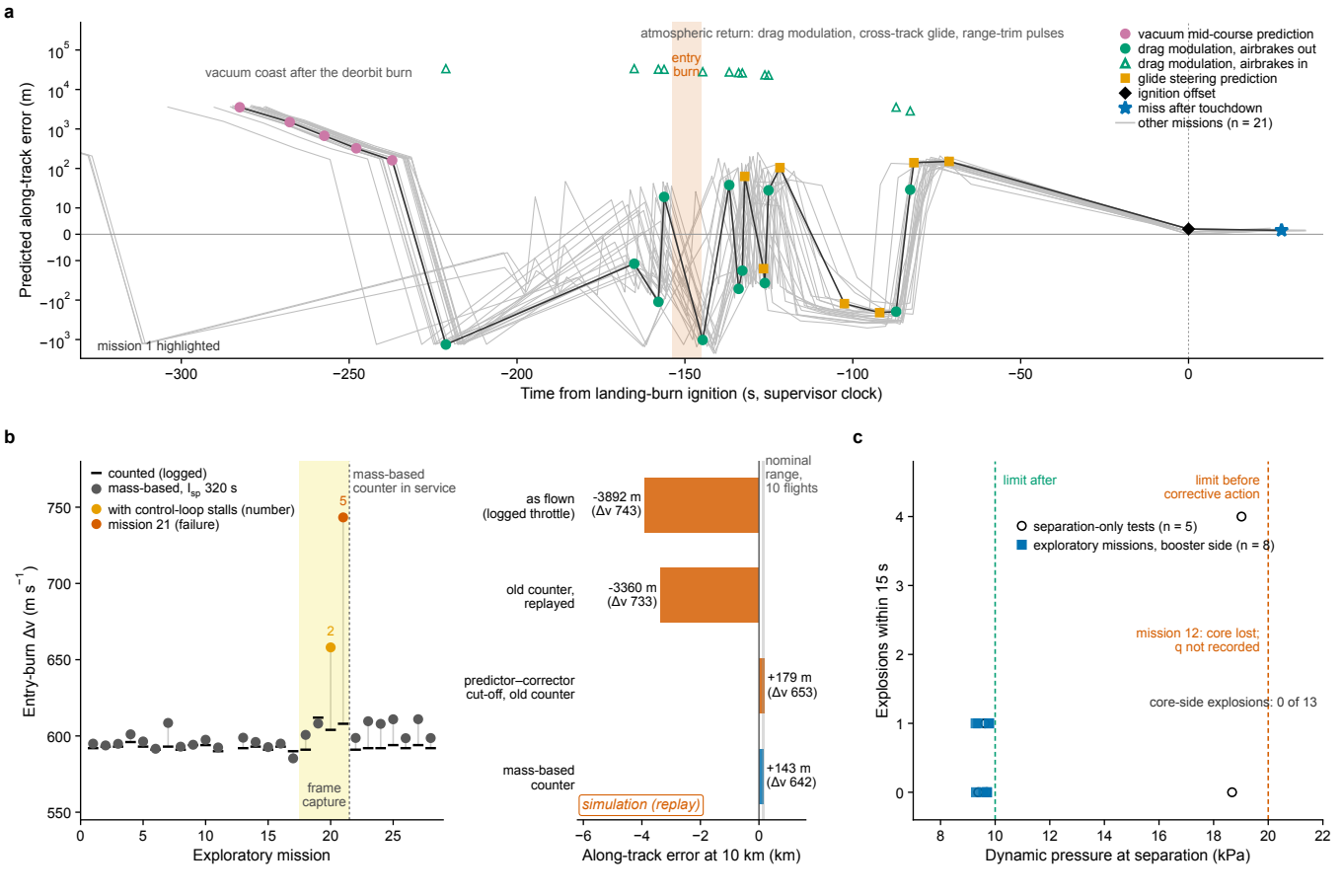

a
atmospheric return: drag modulation, cross-track glide, range-trim pulses
vacuum coast after the deorbit burn
entry burn
vacuum mid-course prediction
drag modulation, airbrakes out
drag modulation, airbrakes in
glide steering prediction
ignition offset
miss after touchdown
other missions (n = 21)
mission 1 highlighted
Predicted along-track error (m)
Time from landing-burn ignition (s, supervisor clock)
b
counted (logged)
mass-based, I_sp 320 s
with control-loop stalls (number)
mission 21 (failure)
mass-based counter in service
frame capture
Entry-burn Δv (m s⁻¹)
Exploratory mission
nominal range, 10 flights
as flown (logged throttle)
-3892 m (Δv 743)
old counter, replayed
-3360 m (Δv 733)
predictor–corrector cut-off, old counter
+179 m (Δv 653)
mass-based counter
+143 m (Δv 642)
simulation (replay)
Along-track error at 10 km (km)
c
limit after
limit before corrective action
separation-only tests (n = 5)
exploratory missions, booster side (n = 8)
mission 12: core lost; q not recorded
core-side explosions: 0 of 13
Explosions within 15 s
Dynamic pressure at separation (kPa)